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Advances in surrogate modeling for storm surge prediction: storm selection and addressing characteristics related to climate change

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Abstract

This paper establishes various advancements for the application of surrogate modeling techniques for storm surge prediction utilizing an existing database of high-fidelity, synthetic storms (tropical cyclones). Kriging, also known as Gaussian process regression, is specifically chosen as the surrogate model in this study. Emphasis is first placed on the storm selection for developing the database of synthetic storms. An adaptive, sequential selection is examined here that iteratively identifies the storm (or multiple storms) that is expected to provide the greatest enhancement of the prediction accuracy when that storm is added into the already available database. Appropriate error statistics are discussed for assessing convergence of this iterative selection, and its performance is compared to the joint probability method with optimal sampling, utilizing the required number of synthetic storms to achieve the same level of accuracy as comparison metric. The impact on risk estimation is also examined. The discussion then moves to adjustments of the surrogate modeling framework to support two implementation issues that might become more relevant due to climate change considerations: future storm intensification and sea level rise (SLR). For storm intensification, the use of the surrogate model for prediction extrapolation is examined. Tuning of the surrogate model characteristics using cross-validation techniques and modification of the tuning to prioritize storms with specific characteristics are proposed, whereas an augmentation of the database with new/additional storms is also considered. With respect to SLR, the recently developed database for the US Army Corps of Engineers' North Atlantic Comprehensive Coastal Study is exploited to demonstrate how surrogate modeling can support predictions that include SLR considerations.

Keywords Kriging · Storm surge · Storm selection · Surrogate model extrapolation · Gaussian process regression · Sea level rise

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1 Introduction

Numerical advances in storm surge prediction over the past couple of decades have produced high-fidelity simulation models that permit a detailed representation of hydrodynamic processes and therefore support high-accuracy forecasting (Luettich et al. 1992; Zijlema 2010). Unfortunately, the computational burden of such numerical models is large, requiring thousands of CPU hours for each simulation (Tanaka et al. 2011), something that limits their applicability for emergency response management and hurricane risk assessment. During landfalling events such models can be utilized to provide only a small number of high-fidelity, deterministic predictions, but cannot support thousand-run ensembles to examine the impact of forecasting errors in the predicted track or provide fast prediction updates once new storm track information becomes available. Their implementation for regional hurricane risk assessment, requiring evaluation of response for a large number of scenarios (Toro et al. 2010), faces similar challenges.

The use of interpolation methodologies and surrogate models (Irish et al. 2009; Das et al. 2010; Jia and Taflanidis 2013; Taflanidis et al. 2013; Kim et al. 2015) has been examined to address the aforementioned challenge. These approaches can provide fast predictions using a database of high-fidelity, synthetic storms, with the goal of maintaining the accuracy of the numerical model utilized to produce this database, while providing greatly enhanced computational efficiency. Leveraging this efficiency they can support emergency response management (Smith et al. 2011) and comprehensive risk assessment (Resio et al. 2009, 2017). Relevant efforts are further promoted by the fact that various such databases are constantly created and updated for regional flooding and coastal hazard studies (Niedoroda et al. 2010; Kennedy et al. 2012; Nadal-Caraballo et al. 2015; USACE 2015). In particular, surrogate models (also frequently referenced as metamodels) are especially attractive for such storm forecasting applications. Formally, surrogate models (metamodels) provide a fast-to-compute mathematical approximation to the input/output relationship of a computationally expensive numerical model, leveraging a database of simulations from this model (synthetic storms for the application examined here). They represent a completely data-driven approach as no explicit information is utilized for the input/output model relationship. This is a critical feature, as it can support any high-fidelity simulation code, any desired parameterization of the synthetic storms and prediction for any output quantities of interest. Kriging, also known as Gaussian process regression, has been demonstrated in a number of studies (Jia and Taflanidis 2013; Jia et al. 2015; Rohmer et al. 2016) to offer great versatility as a surrogate model in this context: (1) It has provided high-accuracy predictions for both storm surge and waves and for either peak or time-series responses, (2) has been efficiently applied for databases of different sizes and characteristics, and (3) has been utilized to provide predictions over large coastal regions, with thousands of save points for which the storm response needs to be evaluated. To enhance computational efficiency for the latter type of implementation, combination with principal component analysis as a dimension reduction technique has been advocated (Jia and Taflanidis 2013). Implementation of Kriging in this greater context (flood risk estimation) has been even recently extended (Bass and Bedient 2018) to address the joint rainfall–runoff and storm surge flood response of tropical cyclones.

This paper offers some important advances for Kriging metamodeling for storm surge prediction applications. In particular, two separate topics are discussed: (a) the exact selection of the synthetic storms for creating the database that will inform the surrogate model development, what is formally known as design of experiments (DoE) in the literature and (b) adjustments for the surrogate model tuning and its implementation to support two issues that might become more relevant due to climate change considerations (Lin et al. 2012), prediction for intensified storms compared to the ones available in initial synthetic-storm database, and surge estimation for sea level rise (SLR) scenarios. A number of computational advances, summarized in the next paragraphs, are established to address these topics with the goals of limiting the number of synthetic-storm simulations, that is, establishing the same surrogate model accuracy using a smaller number of properly chosen storms, or improving prediction accuracy when the Kriging metamodel is used to predict surge responses for storms with different characteristics than the available database (for our application greater storm intensity). These advances are demonstrated here using the recently developed by the US Army Corps of Engineers' database for the North Atlantic Comprehensive Coastal Study (NACCS) (Nadal-Caraballo et al. 2015). Emphasis in all discussions is placed on numerical advances in the tuning of the surrogate model. The demonstration of the subsequent implementation of the (tuned) surrogate model in hindcasting/forecasting or hazard applications falls outside the scope of this study. Fundamental issues related to these implementations, such as addressing uncertainties stemming from the different error sources (including the one from the surrogate model approximation), have been broadly discussed in the relevant literature (Resio et al. 2007, 2009, 2012; Jia and Taflanidis 2013; Taflanidis et al. 2013). Validation of the proposed developments is performed, following current practices in the metamodel literature (Meckesheimer et al. 2002), by evaluating accuracy trends within the database of synthetic storms, rather than restricting comparisons to the implementation of the surrogate model to hindcast/forecast specific storms.

Related to the first established advancement, the selection of synthetic storms, surrogate modeling applications have used so far existing databases of synthetic storms, established using, typically, the joint probability method with optimal sampling (JPM-OS) (Toro et al. 2010) or, less frequently, orthogonal DoE (Kennedy et al. 2012). An adaptive, sequential DoE is examined here that iteratively identifies the storm (or multiple storms) that is expected to provide the greatest enhancement of the prediction accuracy when that storm is added into the database. The overall objective of this adaptive storm selection is to establish a high-accuracy surrogate model using the smallest possible number of synthetic-storm simulations, thus minimizing the associated computational burden. This is accomplished by iteratively performing new storm simulations, using the existing Kriging metamodel to guide the selection of these new storms. As the size of database increases, the surrogate model is re-tuned and its accuracy is re-evaluated. The iterative process stops when that accuracy reaches a predetermined threshold, considered sufficient for facilitating high-quality predictions. The sufficiency, that is, the accuracy of the surrogate model supported predictions, and efficiency, that is, the number of storms needed to establish a satisfactory accuracy, of the adaptive storm selection are compared to the selection of storms using JPM-OS.

For addressing storm intensification, the use of the surrogate model for prediction extrapolation is discussed. Such extrapolation might be necessitated for predictions of storms of (higher) intensity outside the parameter space of the existing database. The tuning of the model characteristics using cross-validation (CV) is utilized for facilitating higher-accuracy extrapolations. Tuning refers here to the selection of the surrogate model hyper-parameters, a task for which maximum likelihood estimation (MLE) has been considered in all previous applications of Kriging for storm surge predictions. Both CV and MLE are additionally modified to allow the incorporation of weights for prioritizing storms with specific characteristics during the surrogate model tuning. For further improvement of extrapolation capabilities, a DoE enrichment strategy is also considered, applicable to cases where new synthetic storms can be included in the database. The overall objective through these developments is to reduce the surrogate model error when it is used for predictions of storms with greater intensity compared to the database of synthetic storms used in its development, either through appropriate tuning (with this objective in mind) or through addition in the database of a small number of, properly selected, new synthetic storms. Finally, the NACCS database is exploited to demonstrate how Kriging surrogate modeling can support predictions that include SLR considerations.

In the next section, all theoretical details of the surrogate modeling framework are presented. This includes an overview of the framework as well as introduction of the advancements established in this work. Discussion focuses on essential theoretical components, with further computational and numerical details presented in separate appendices. Section 3 offers an overview of the database used in this study. The storm selection is discussed in Sect. 4, and implications related to climate change characteristics (storm intensification and sea level rise) are presented in Sect. 5. Finally, Sect. 6 offers concluding remarks.

2 Overview of Kriging surrogate modeling and theoretical advances

This section discusses all the theoretical advances established in this paper for Kriging metamodeling applied to storm surge prediction. Emphasis is placed on the essential mathematical aspects of the surrogate model development, needed to support the relevant advances. A more detailed discussion on the implementation of Kriging in this context (storm surge prediction) is included in (Jia et al. 2015).

2.1 Fundamentals for Kriging implementation for storm surge forecasting

For establishing the surrogate model, an appropriate parameterization of the synthetic storms is required to provide the model input. Typically, characteristics close to landfall are selected for this parameterization, assuming that variability prior to landfall is addressed by appropriate selection of each track history (Resio et al. 2009). Note that this is the same approach utilized by the joint probability method (JPM), the conventional methodology for calculating storm risk (Toro et al. 2010). Therefore, the necessary parameterization for defining the surrogate model input can be easily facilitated for databases developed using JPM principles.

Based on the adopted parameterization, each synthetic storm is characterized by n_x-dimensional input vector $\mathbf{x} \in \Re^{n_x}$ and provides predictions for the n_y-dimensional output vector $\mathbf{y}(\mathbf{x})$. The different components of $\mathbf{y}(\mathbf{x})$, denoted $y_j(\mathbf{x})$, may pertain to a response output at a specific coastal location or specific instance in time (for time-dependent predictions (Jia et al. 2015)). Let n be the number of available synthetic storms. The database includes storm scenarios $\{\mathbf{x}^h;\ h = 1,\ldots,\ n\}$, frequently referenced as training points or experiments in surrogate modeling terminology, associated with output predictions $\{\mathbf{y}^h;\ h = 1,\ldots,n\}$, frequently referenced as observations. We will use $\mathbf{X} = [\mathbf{x}^1 \quad \ldots \quad \mathbf{x}^n]^{\mathrm{T}} \in \Re^{n \times n_x}$ and $\mathbf{Y} = [\mathbf{y}^1 \quad \ldots \quad \mathbf{y}^n]^{\mathrm{T}} \in \Re^{n \times n_y}$ to denote the corresponding input and output matrices, respectively, and X to denote the input domain of the training storms. The latter is defined for each input component through the interval $\left[x_k^{\min}\ x_k^{\max}\right]$, where $x_k^{\min}$ and $x_k^{\max}$ are, respectively, the minimum and maximum values considered for input $\left\{x_k^h; h = 1, \ldots, n\right\}$.

For implementation over an extended coastal region the dimension of n_y can be very large, which poses challenges in terms of computational efficiency. In such cases the adoption of principal component analysis (PCA) as a dimensional reduction technique has been shown to be beneficial (Jia and Taflanidis 2013). PCA provides a low-dimensional space of m_c latent outputs ($m_c \ll n_y$), corresponding to the eigenvectors associated with the m_c largest eigenvalues of the covariance matrix for observations $\mathbf{Y}$, and the surrogate model can be easier developed in this space. The predictions are first established for the latent output vector $\mathbf{z}$ (with each component denoted herein as z_i) and then transformed back to the initial, high-dimensional output vector $\mathbf{y}$. The relationship between $\mathbf{y}$ and $\mathbf{z}$ is established through linear transformation using the projection matrix $\mathbf{P}$, corresponding to the aforementioned retained eigenvectors. The exact value of m_c may be chosen so that the associated observation $\mathbf{Z}$ for output $\mathbf{z}$ accounts for significant portion of the variability (for example, 99.9%) of the initial observation matrix $\mathbf{Y}$, or by performing a sensitivity analysis for the relative benefits (Jia and Taflanidis 2013), examining the trade-off between computational efficiency and accuracy.

The surrogate model is established for vector $\mathbf{z}$ with observation matrix $\mathbf{Z}$ (or $\mathbf{y}$ and $\mathbf{Y}$, respectively, if PCA not used). Kriging approximates the true response as one realization of a Gaussian process that also includes a regression model. Its fundamental building blocks are the n_p-dimensional basis vector, $\mathbf{f}(\mathbf{x})$, and the correlation function $R(\mathbf{x}^l,\mathbf{x}^m|\mathbf{s})$ which is dependent on hyper-parameters $\mathbf{s}$. The former represents the underlying regression model, typically chosen as lower-order polynomial, and the latter the correlation for the Gaussian process. A common choice for this function is the generalized exponential correlation:

$$R\left(\mathbf{x}^l, \mathbf{x}^m|\mathbf{s}\right) = \prod_{k=1}^{n_x} \exp\left[-s_k|x_k^l - x_k^m|^{s_{n_x+1}}\right]; \mathbf{s} = [s_1 \ \cdots \ s_{n_x+1}]. \tag{1}$$

For the set of n observations with input matrix $\mathbf{X}$ and corresponding output matrix $\mathbf{Z}$, we then define the $n \times n_p$ basis matrix $\mathbf{F}(\mathbf{X}) = [\mathbf{f}(\mathbf{x}^1)\ldots\mathbf{f}(\mathbf{x}^n)]^{\mathrm{T}}$ and the $n \times n$ correlation matrix $\mathbf{R}(\mathbf{X})$ with the lm-element defined as $R(\mathbf{x}^l,\mathbf{x}^m|\mathbf{s})$, $l,m = 1, \ldots, n$. Also for every new input $\mathbf{x}$ we define the n-dimensional correlation vector $\mathbf{r}(\mathbf{x}|\mathbf{X}) = [R(\mathbf{x},\mathbf{x}^1|\mathbf{s}) \ \ldots \ R(\mathbf{x},\mathbf{x}^n|\mathbf{s})]^{\mathrm{T}}$ between the input and each of the elements of $\mathbf{X}$. The Kriging prediction (given as row vector) is finally (Sacks et al. 1989)

$$\hat{\mathbf{z}}(\mathbf{x}|\mathbf{X}) = \mathbf{f}(\mathbf{x})^{\mathrm{T}}\boldsymbol{\beta}^* + \mathbf{r}(\mathbf{x}|\mathbf{X})^{\mathrm{T}}\mathbf{R}(\mathbf{X})^{-1}(\mathbf{Z} - \mathbf{F}(\mathbf{X})\boldsymbol{\beta}^*), \tag{2}$$

where $\boldsymbol{\beta}^* = \left(\mathbf{F}(\mathbf{X})^{\mathrm{T}}\mathbf{R}(\mathbf{X})^{-1}\mathbf{F}(\mathbf{X})\right)^{-1}\mathbf{F}(\mathbf{X})^{\mathrm{T}}\mathbf{R}(\mathbf{X})^{-1}\mathbf{Z}$. These predictions can be then transformed to predictions for output $\hat{\mathbf{y}}(\mathbf{x}|\mathbf{X})$ through the PCA transformation (Jia et al. 2015). In addition, Kriging provides an estimate of the accuracy of this prediction, termed the predictive variance, which is a function of $\mathbf{x}$, and for output component z_i is (Sacks et al. 1989)

$$\begin{aligned}\sigma_i^2(\mathbf{x}|\mathbf{X}) &= \tilde{\sigma}_i^2(\mathbf{X})\left[1 + \mathbf{u}(\mathbf{x}|\mathbf{X})^{\mathrm{T}}\left\{\mathbf{F}(\mathbf{X})^{\mathrm{T}}\mathbf{R}(\mathbf{X})^{-1}\mathbf{F}(\mathbf{X})\right\}^{-1}\mathbf{u}(\mathbf{x}|\mathbf{X}) - \mathbf{r}(\mathbf{x}|\mathbf{X})^{\mathrm{T}}\mathbf{R}(\mathbf{X})^{-1}\mathbf{r}(\mathbf{x}|\mathbf{X})\right] \\ &= \tilde{\sigma}_i^2(\mathbf{X})\sigma_n^2(\mathbf{x}|\mathbf{X}),\end{aligned} \tag{3}$$

where $\mathbf{u}(\mathbf{x}|\mathbf{X}) = \mathbf{F}(\mathbf{X})^{\mathrm{T}}\mathbf{R}(\mathbf{X})^{-1}\mathbf{r}(\mathbf{x}|\mathbf{X}) - \mathbf{f}(\mathbf{x})$ and the process variance $\tilde{\sigma}_i^2(\mathbf{X})$ is given by

$$\tilde{\sigma}_i^2(\mathbf{X}) = \frac{1}{n}\sum_{h=1}^{n}[\boldsymbol{\rho}(\mathbf{X})]_{hi}^2, \tag{4}$$

with notation $[.]_{pq}$ used herein to denote the entry on the pth row and qth column in a matrix (i.e., pqth element of the matrix), and matrix $\boldsymbol{\rho}(\mathbf{X}) = \mathbf{C}(\mathbf{X})^{-1}(\mathbf{Z} - \mathbf{F}(\mathbf{X})\boldsymbol{\beta}^*)$, where $\mathbf{C}(\mathbf{X})$ corresponds to the lower Cholesky factorization for $\mathbf{R}(\mathbf{X})$. Each of the columns of the matrix $\boldsymbol{\rho}(\mathbf{X})$ represents the weighted regression error (across the different storms) for a specific output. Note that the variation of $\sigma_i(\mathbf{x}|\mathbf{X})$ with respect to $\mathbf{x}$ is determined by the quantity in the brackets in Eq. (3), termed herein normalized variance $\sigma_n^2(\mathbf{x}|\mathbf{X})$, which is independent of the observations $\mathbf{Y}$, and therefore common for all outputs. The difference between the outputs stems from $\tilde{\sigma}_i^2(\mathbf{X})$, which is simply a scaling factor. Approaches for validation of the surrogate model are discussed in Appendix 1.

2.2 Tuning of surrogate model hyper-parameters

The optimal selection (tuning) of the hyper-parameter vector $\mathbf{s}$ is critical for prediction accuracy. This tuning is typically performed through the maximum likelihood estimation (MLE) principle, where the likelihood, that is, the objective function in the hyper-parameter optimization, is defined as the probability of the n observations. Maximizing this likelihood with respect to $\mathbf{s}$ ultimately leads to

$$\mathbf{s}^* = \arg\min_{\mathbf{s}} \left[|\mathbf{R}(\mathbf{X})|^{\frac{1}{n}} \sum_{i=1}^{m_c} \gamma_i \tilde{\sigma}_i^2 \right], \tag{5}$$

where |.| stands for determinant of a matrix and γ_i is a weight for each output quantity, representing its relative weight. Standard approaches for solving this optimization are given in Lophaven et al. (2002). In this paper, the gradient-based optimization discussed in (Lophaven et al. 2002) is combined with a global random search (as a first step) to avoid convergence to suboptimal solutions since it is well established (Santner et al. 2013) that optimization of Eq. (5) has multiple local minima.

So far for all applications of Kriging for storm surge forecasting the MLE approach has been considered for the hyper-parameter tuning. Here an alternative approach is examined, termed CV, the tuning based on the surrogate model predictive capabilities (Sundararajan and Keerthi 2001), meaning its ability to provide accurate estimates for storms outside of the database used to develop it. Any validation approach, like the ones discussed in Appendix 1, can be used for selecting the accuracy statistics for definition of the objective function. The most convenient and popular implementation is leave-one-out cross-validation (LOOCV) leading to objective function definition

$$H_m(\mathbf{s}) = \frac{1}{m_c} \sum_{i=1}^{m_c} \gamma_i \left[\frac{1}{n} \sum_{h=1}^{n} e_{hi}^2 \right]; \quad e_{hi} = z_i\left(\mathbf{x}^h\right) - \hat{z}_i\left(\mathbf{x}^h | \mathbf{X}_{-h}\right), \tag{6}$$

where the quantity in the brackets corresponds to the leave-one-out (LOO) error for output z_i and $\hat{z}_i(\mathbf{x}^h|\mathbf{X}_{-h})$ represents the surrogate model predictions for storm $\mathbf{x}^h$ established by using the entire database excluding that storm. As discussed in Appendix 1, the above formulation of LOOCV is established by sequentially removing each storm from the database, using the remaining storms as training points to predict the output for that storm, and finally calculating the discrepancy between the predicted and real responses.

Actually, a closed form approximation exists for e_{hi} with no need to evaluate $\hat{z}_i\left(\mathbf{x}^h|\mathbf{X}_{-h}\right)$ (Sundararajan and Keerthi 2001):

$$e_{hi} = \frac{\left[\mathbf{R}(\mathbf{X})^{-1}(\mathbf{Z} - \mathbf{F}(\mathbf{X})\boldsymbol{\beta}^*)\right]_{hi}}{\left[\mathbf{R}(\mathbf{X})^{-1}\right]_{hh}} = \frac{g_{hi}}{c_{hh}} \tag{7}$$

where as shown above, c_{hh} corresponds to the hth diagonal entry of $\mathbf{R}(\mathbf{X})^{-1}$ and g_{hi} corresponds to the hith element of matrix $\mathbf{g} = \mathbf{R}(\mathbf{X})^{-1}(\mathbf{Z} - \mathbf{F}(\mathbf{X})\boldsymbol{\beta}^*)$. This analytic expression for e_{hi} is a major reason for the popularity of LOOCV as objective for hyper-parameter tuning. This tuning corresponds ultimately to optimization

$$\mathbf{s}^* = \arg\min_{\mathbf{s}} H_m(\mathbf{s}) \tag{8}$$

which can be solved through any appropriate numerical scheme (Sundararajan and Keerthi 2001). The calculation of the gradient of the objective, which can improve efficiency of the optimization, is discussed in Appendix 2. Similar to the MLE case, a global random search is also used for the LOOCV tuning (as a first step) to avoid convergence to local suboptimal solutions.

2.3 Tuning of hyper-parameter with preference to storms with specific characteristics

The hyper-parameter tuning approaches discussed in the previous subsection give an equal weight across all the storms in the database. It might be beneficial, though, to give higher priority to storms within the database that have some specific characteristics. This will be desired, for example, when the developed surrogate model will be required to provide predictions for storms with higher intensity than the ones considered within the available database. In this case, prioritizing higher-intensity storms within the database is anticipated to provide improved prediction accuracy for storms with even greater intensity.

This prioritization of different storms within the database can be accommodated by introducing weight w_h for each storm and by using a weighted average, rather than simply an average, for calculating the desired statistics across the database. For the MLE tuning this requires substitution of $\tilde{\sigma}_i^2$ in Eq. (4) with

$$ws_i^2 = \frac{1}{\sum_{h=1}^{n} w_h} \sum_{h=1}^{n} w_h [\boldsymbol{\rho}(\mathbf{X})]_{hi}^2 \tag{9}$$

For the LOOCV tuning this requires modification of $H_m(\mathbf{s})$ to

$$H_m^w(\mathbf{s}) = \frac{1}{m_c} \sum_{i=1}^{m_c} \gamma_i \left[\frac{1}{\sum_{h=1}^{n} w_h} \sum_{h=1}^{n} w_h e_{hi}^2\right] \tag{10}$$

2.4 Storm selection for formulation of the database

The accuracy of Kriging is significantly influenced by the proper selection of the training points (DoE). As discussed in the introduction, studies that have examined application of Kriging for storm surge predictions have been constrained to existing databases, developed for coastal flood studies using typically JPM-OS (Toro et al. 2010). An alternative approach would be to rely on a sequential, adaptive DoE (Sacks et al. 1989; Liu et al. 2017) that supports a gradual enrichment of the database that informs the surrogate model development. Note that the broader topic of selection of storms for coastal risk assessment applications has received significant attention (Toro et al. 2010; Fischbach et al. 2016). The adaptive DoE discussed here, though, approaches this topic from a fundamentally different perspective, as it investigates the selection of synthetic storms to optimally inform the development of a Kriging metamodel, not simply for the estimation of surge hazard for a specific probabilistic description of future storms. The surrogate model tuned using the current database of synthetic storms is explicitly utilized, specifically its predictive variance, to select the next storm.

Given the database $\mathbf{X}$ and an already tuned surrogate model, the adaptive DoE searches for the new experiment (storm in our application) $\mathbf{x}_{new}$ that will provide the best anticipated improvement of accuracy based on some error measure. The normalized, weighted integrated mean squared error, *IMSE*, is chosen here (Sacks et al. 1989) for error measure given by

$$IMSE(\mathbf{X}, \mathbf{x}_{new}) = \int_{X^I} \varphi(\mathbf{x}) \sigma_n^2(\mathbf{x}|\mathbf{X}, \mathbf{x}_{new}) \mathrm{d}\mathbf{x} \tag{11}$$

where X^I is the input domain of interest, $\varphi(\mathbf{x})$ the *IMSE* weight function, and $\sigma_n^2(\mathbf{x}|\mathbf{X}, \mathbf{x}_{new})$ the normalized predictive variance considering as experiments $\mathbf{X}$ and $\mathbf{x}_{new}$, given by

$$\begin{aligned}\sigma_n^2(\mathbf{x}|\mathbf{X}, \mathbf{x}_{new}) = 1 + \mathbf{u}(\mathbf{x}|\mathbf{X}, \mathbf{x}_{new})^{\mathrm{T}} \left\{ \mathbf{F}(\mathbf{X}, \mathbf{x}_{new})^{\mathrm{T}} \mathbf{R}(\mathbf{X}, \mathbf{x}_{new})^{-1} \mathbf{F}(\mathbf{X}, \mathbf{x}_{new}) \right\}^{-1} \mathbf{u}(\mathbf{x}|\mathbf{X}, \mathbf{x}_{new}) \\ - \mathbf{r}(\mathbf{x}|\mathbf{X}, \mathbf{x}_{new})^{T} \mathbf{R}(\mathbf{X}, \mathbf{x}_{new})^{-1} \mathbf{r}(\mathbf{x}|\mathbf{X}, \mathbf{x}_{new})\end{aligned} \tag{12}$$

Note that this normalized variance does not correspond to a specific output, that is, it ignores scaling $\tilde{\sigma}_i^2$, since the latter has no impact on the optimization examined here. The term $IMSE(\mathbf{X}, \mathbf{x}_{new})$ corresponds to the average (across domain X^I) error of the surrogate model established by using support points $\mathbf{X}$ and $\mathbf{x}_{new}$, assuming no modification in the hyper-parameters after the addition of $\mathbf{x}_{new}$. X^I should be chosen as the input domain in which we are interested in controlling the prediction error, whereas the function $\varphi(\mathbf{x})$ can be used to give higher priority to specific subdomains. Selection of $\varphi(\mathbf{x})$ depends on the application of interest. If Kriging is intended to be used to provide predictions for arbitrary storms, then no preference exists and $\varphi(\mathbf{x})$ should correspond to a uniform distribution. This might be the case, for example, when developing a surrogate model to manage emergency responses during landfalling events (Kijewski-Correa et al. 2014). If, on the other hand, Kriging is intended to provide predictions for specific, known storms, then $\varphi(\mathbf{x})$ should be chosen to reflect this. This might be the case, when surrogate model is intended to provide risk estimates for specific domain and JPM distribution is known.

The evaluation of $\sigma_n^2(\mathbf{x}|\mathbf{X}, \mathbf{x}_{new})$ requires ultimately the estimation of the correlation matrix $\mathbf{R}(\mathbf{X}, \mathbf{x}_{new})$, the basis function vector $\mathbf{F}(\mathbf{X}, \mathbf{x}_{new})$ and the correlation vector $\mathbf{r}(\mathbf{x}|\mathbf{X}, \mathbf{x}_{new})$ considering training points as the augmentation of both the existing ones $\mathbf{X}$ and the potential new one(s) $\mathbf{x}_{new}$. It does not involve, though, the storm output for $\mathbf{x}_{new}$ which is why *IMSE* can be calculated by knowing simply the new experiment input. Estimation of $\sigma_n^2(\mathbf{x}|\mathbf{X}, \mathbf{x}_{new})$ requires inversion of the updated correlation matrix,

$\mathbf{R}(\mathbf{X},\mathbf{x}_{new})^{-1}$, which can be computationally demanding. This inversion can be substituted by equivalent expression (Sacks et al. 1989)

$$\mathbf{R}(\mathbf{X},\mathbf{x}_{new})^{-1} = \begin{bmatrix} \mathbf{R}(\mathbf{X})^{-1} + \frac{1}{\eta_{new}}\mathbf{R}(\mathbf{X})^{-1}\mathbf{r}(\mathbf{x}_{new}|\mathbf{X})\mathbf{r}(\mathbf{x}_{new}|\mathbf{X})^{\mathrm{T}}\mathbf{R}(\mathbf{X})^{-1} & -\frac{1}{\eta_{new}}\mathbf{R}(\mathbf{X})^{-1}\mathbf{r}(\mathbf{x}_{new}|\mathbf{X}) \\ -\frac{1}{\eta_{new}}\mathbf{r}(\mathbf{x}_{new}|\mathbf{X})^{\mathrm{T}}\mathbf{R}(\mathbf{X})^{-1} & \frac{1}{\eta_{new}} \end{bmatrix} \tag{13}$$

where $\eta_{new} = 1 - \mathbf{r}(\mathbf{x}_{new}|\mathbf{X})^{\mathrm{T}}\mathbf{R}(\mathbf{X})^{-1}\mathbf{r}(\mathbf{x}_{new}|\mathbf{X})$. This expression does not entail any new matrix inversions.

The new training point(s) can be finally selected to minimize *IMSE* as

$$\mathbf{x}_{new} = \arg\min_{\mathbf{x}_{new}\in D} IMSE(\mathbf{X},\mathbf{x}_{new}) \tag{14}$$

where D is the candidate domain for $\mathbf{x}_{new}$, for example X^I. This optimization is well-posed (always has a solution), is known to have multiple local minima, and could be computationally expensive since it involves an integration to calculate $IMSE(\mathbf{X},\mathbf{x}_{new})$. Stochastic principles are adopted here to address both these challenges: The optimization of Eq. (14) is performed through means of stochastic search considering n_c candidates for $\mathbf{x}_{new}$, whereas the integral of Eq. (11) is estimated through Monte Carlo integration (MCI) considering N_s samples of $\mathbf{x}$ from $\varphi(\mathbf{x})$. A detailed optimization scheme is discussed in Appendix 3.

Optimization of Eq. (14) ultimately identifies the experiment (storm for our application) that is expected to provide the biggest accuracy improvement measured by the reduction of *IMSE*. This is demonstrated in Fig. 1 for a two-dimensional example. The output in this example corresponds to the storm surge at a coastal location in New Jersey (DD coordinates 38.9532, − 74.8503). This surge is estimated through the NACCS synthetic-storm database presented in the next section. To facilitate here an illustrative two-dimensional example, the varied inputs are only the central pressure deficit ΔP and the forward speed v_f with the remaining storm characteristics constrained to specific values x_{lat} = 35.52°N, x_{lon} = −74.09°W, θ = − 40°, and R_m = 43 km (for proper definition of the storm parameters see Sect. 3). This ultimately represents a simplification, for the purposes of providing a two-dimensional illustration with respect to a scalar output, of the storm selection problem that will be examined in Sect. 4. Part (a) of this figure shows the surge response and its Kriging approximation for database $\mathbf{X}$ consisting of 5 synthetic storms, whereas part b the initial normalized variance $\sigma_n^2(\mathbf{x}|\mathbf{X})$. Parts (c)–(e) then show the updated $\sigma_n^2(\mathbf{x}|\mathbf{X},\mathbf{x}_{new}^c)$ for three different candidate experiments $\mathbf{x}_{new}^c; c = 1,\ldots,3$. In all cases, the *IMSE* is also reported. The third experiment $\mathbf{x}_{new}^3$ is the one that provides the greater reduction in *IMSE* and therefore should be chosen. Finally, part (f) shows the updated variance when addition of another experiment is considered, beyond the first optimal experiment $\mathbf{x}_{new}^3$.

Once $\mathbf{x}_{new}$ is identified, it is added in the database, its output $\mathbf{y}(\mathbf{x}_{new})$ is evaluated and a new Kriging model is formulated (hyper-parameter re-tuning). Alternatively, a batch of new storms may be identified before updating, that is re-tuning, the surrogate model. After each new storm is identified it is added in the database, but the hyper-parameters are not always updated. The normalized variance is simply updated by including the identified

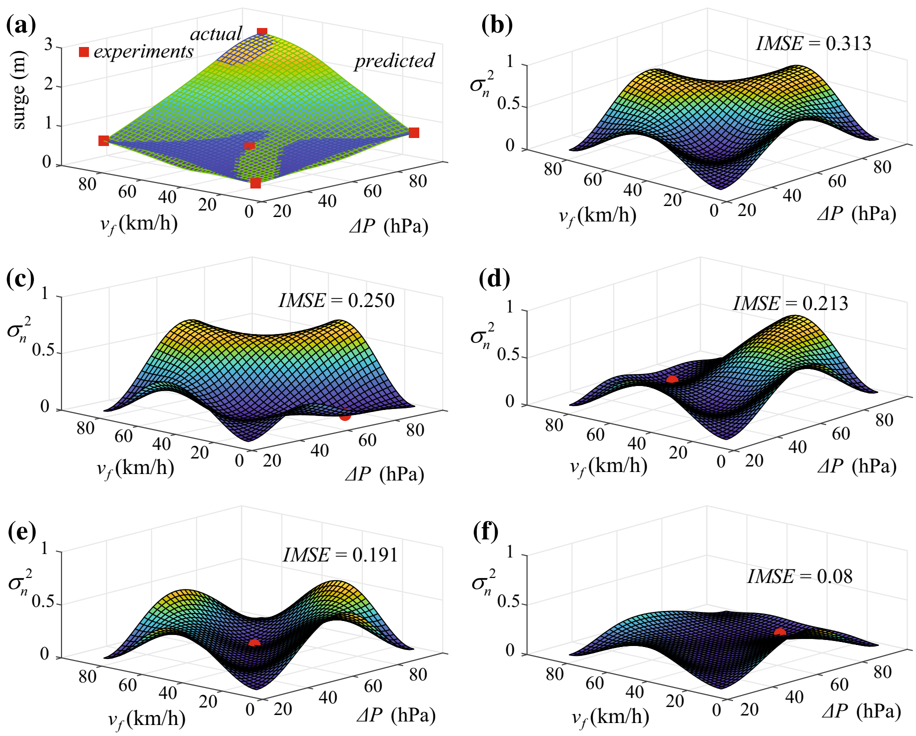

Fig. 1 Demonstration of adaptive DoE for a two-dimensional example corresponding to storm surge at a coastal location in New Jersey (DD coordinates 38.9532, − 74.8503) as a function of central pressure deficit ΔP and forward speed v_f. **a** Actual surge response and Kriging approximation and **b** initial and four different (**c**)–(**f**) updated normalized variances. Updates correspond to the addition of a different new experiment in each case, or of two experiments in part **f**. The *IMSE* is also reported for each case

experiment $\mathbf{x}_{new}$ in $\mathbf{X}$. This process moves iteratively to identification of the next storm, and only when the required number of batch storms is obtained, the surrogate model's hyper-parameters are updated. Parts (e), (f) of Fig. 1 illustrate this concept. This sequential addition of storms, performed either one at a time or through batches, terminates when the updated surrogate model satisfies desired accuracy criteria (has sufficient accuracy), or the number of storm simulations exceeds allowable computational budget. Advanced information-theoretic termination criteria (Kleijnen and Beers 2004) can be also utilized, that attempt to establish a balance between the benefits of the accuracy enhancement offered by adding new storms in the database and the computational burden of performing the additional simulations for these storms. These termination criteria will be ultimately application dependent, taking the available computational resources and the intended use of the surrogate model into account.

Note that instead of integrated mean squared error, the maximum (Sacks et al. 1989) mean squared error (MMSE) could have been used for the DoE, given by substituting the integration in Eq. (11) with the maximum of the integrand, corresponding to the maximum error over domain X^I. However, for continuous input domains such MMSE criterion is not recommended as it is usually highly multimodal and challenging to optimize (Sacks et al. 1989). Another approach would have been to incorporate considerations about the LOO error (Liu et al. 2017) in the formulation of the objective for identification of the next optimal experiment. However, such approach is typically applied to scalar outputs; extensions to a high-dimensional output case, like the one examined here, are uncommon. Other closely related DoE approaches, such as the maximum entropy approach (Wynn 2004), rely on information-theoretic measures and thus lack compatibility to aforementioned weights. Interested readers are welcomed to refer to Pronzato and Müller (2012) for details about recent development on various DoE approaches.

3 NACCS database details

The database utilized in this study corresponds to the greater US North Atlantic coast (Virginia to Maine) and was developed by the US Army Corps of Engineers for supporting the North Atlantic Comprehensive Coastal Study (NACCS) (Nadal-Caraballo et al. 2015; USACE 2015). The NACCS tropical cyclone database consists of a total of 1050 synthetic storms. However, for the purpose of this study, where storm surge-only simulation results are employed, that is, excluding astronomical tide, quality assurance checks reduce this database to 1031 storms with validated results, separated into two clusters, 595 landfalling storms and 436 bypassing storms. These two clusters have different characteristics, and separate surrogate model development should be considered for each. Herein focus will be on the landfalling storms, since their larger number facilitates a higher prediction accuracy. Storm surge was simulated by a coupled high-fidelity numerical hydrodynamic model composed of STWAVE [steady-state spectral wave model] (Smith et al. 2001) and ADCIRC [advanced circulation model] (Luettich et al. 1992), with the planetary boundary layer (PBL) model providing wind and pressure input to this coupled hydrodynamic model. The computational domain for generating the synthetic storms consists of 3.1 million computational nodes and 6.2 million elements and encompasses the western North Atlantic, the Gulf of Mexico and the western extent of the Caribbean Sea. The largest elements are in the Caribbean Sea, with nodal spacing of about 40 km. The smallest elements resolve detailed geographic features such as tributaries, where nodal spacing is approximately 10 m. Further details are discussed in Nadal-Caraballo et al. (2015).

The synthetic storms were generated utilizing the JPM-OS approach considering the historic characteristics of regional storms (probable combinations of storm intensity as well as track and landfalling location for the region), with the input vector composed of: the latitude, x_{lat}, and longitude, x_{lon}, of the landfall location, the heading direction θ at that location, the central pressure deficit ΔP, the forward speed v_f, and the radius of maximum winds R_m. The latter three are determined offshore, when the storm has its maximum intensity (this intensity starts to reduce close to landfall (Nadal-Caraballo et al. 2015)). Therefore, $\mathbf{x} = [x_{lat}\ x_{lon}\ \theta\ \Delta P\ v_f\ R_m]$ with $n_x = 6$. Eighty-nine different master tracks, shown in Fig. 2, were considered for the landfalling storms. All master tracks were varied with different spacing across different subregions of the North Atlantic to create the total number of storm tracks. Landfalling storms have track headings of − 60°, − 40°, − 20°, and 0° (clockwise from north) at the point of landfall. The parameter ranges and values for the remaining inputs are the ones shown the histogram of Fig. 3.

Discussion is also going to focus on estimation of risk for two representative locations, corresponding to The Battery, New York (DD coordinates 40.7000, − 74.0133), and Virginia Beach, VA (DD coordinates 36.8450, − 75.9352). The benchmark risk for these locations is obtained by utilizing the entire suite of the 595 NACCS landfalling storms based on their respective JPM-OS weights λ_j^l, where λ_j^l denotes the frequency weight (mean annual rate) for landfalling storm l for save location j. This risk will be represented as

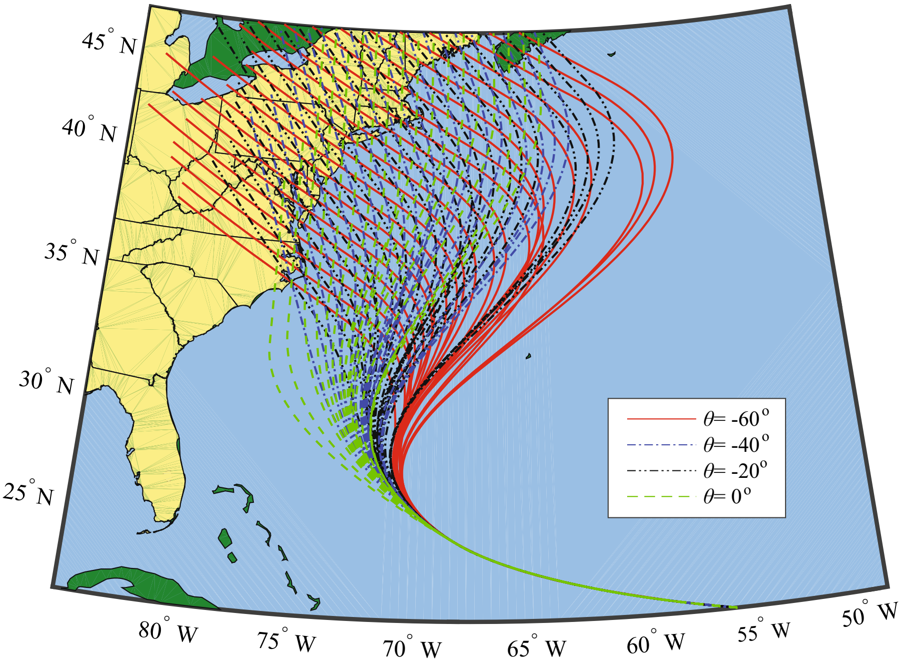

Fig. 2 Storm tracks for the landfalling synthetic storms

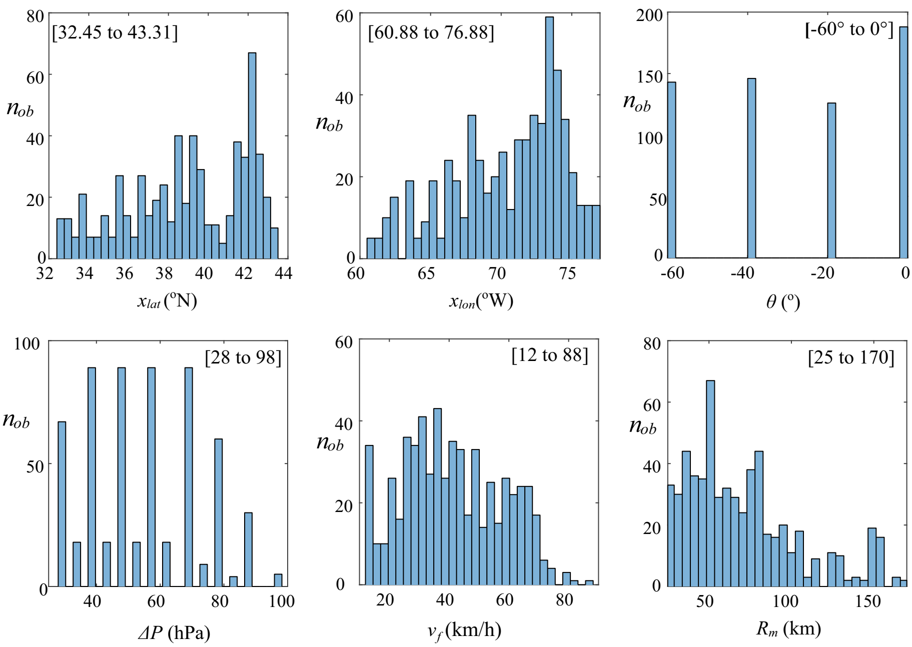

Fig. 3 Histogram of the input parameter values for landfalling storms in the NACCS database. For each input x_k the range X_k is also reported in brackets

annual exceedance rate $\lambda_j(\beta)$ that the surge (due to landfalling storms) will exceed threshold β and is calculated as

$$\lambda_j(\beta) = P[y_j > \beta | 1 \ \text{year}] = \frac{1}{n_l} \sum_{l=1}^{n_l} \lambda_j^l P\left[y_j(\mathbf{x}^l) > \beta\right] \tag{15}$$

where $\{\mathbf{x}^l; l = 1,\ldots,n_l\}$ corresponds to the set of n_l = 595 landfalling storms in the NACCS database.

4 Storm selection

The first topic investigated corresponds to the efficiency/sufficiency of the storm database for supporting the surrogate model development. The number of synthetic storms n for forming the database is gradually increased by selecting new storms to add, and for each n value, a new Kriging model is optimized using CV approach. Preference for CV will be justified in the next section. The accuracy of the Kriging model is then evaluated using LOOCV as well as test-sample approach.

Four different approaches are examined for the DoE, summarized in Table 1. These are also illustrated in Fig. 4 looking at projection to ΔP and R_m inputs. That figure also includes the different definitions of weights for *IMSE* calculation that will be discussed in the next paragraph. The first approach, termed JPM-RnD, randomly picks one storm from the 595 available JPM-OS storms forming the NACCS database. The remaining three approaches consider the *IMSE*-based optimization discussed in Sect. 2.4, with different definitions for D. The second approach, termed JPM-Optimal, simply picks sequentially the best storm from the 595 available JPM-OS storms forming the NACCS database. In this case, D for the optimization of Eq. (14) is constrained to the 595 storms. The surrogate model is updated, and its accuracy is re-evaluated after addition of 10 new storms. The third and fourth approaches consider the best storm in D, with the candidate domain D defined as an extension of X, corresponding to a 20% increase

Table 1 Summary of the characteristics of the different DoE approaches examined

Name	Description
Storm selection approach	
DoE10	Adaptive selection of "best" storms to add by optimization over a continuous domain. Kriging hyper-parameters are updated once 10 new storms are added
DoE80	Same as DoE10, with the only difference being that the Kriging hyper-parameters are updated once 80 new storms are added
JPM-Optimal	Same as DoE10, with the only difference being that the "best" storm is sought among the 595 JPM-OS storms (not continuous domain)
JPM-RnD	Baseline method. The added storm is randomly picked from the available 595 JPM-OS storms
Weight selection for *IMSE* definition	
Uniform DoE prioritization	Weights are uniform over the entire domain
JPM DoE prioritization	Weights correspond to JPM-OS distribution

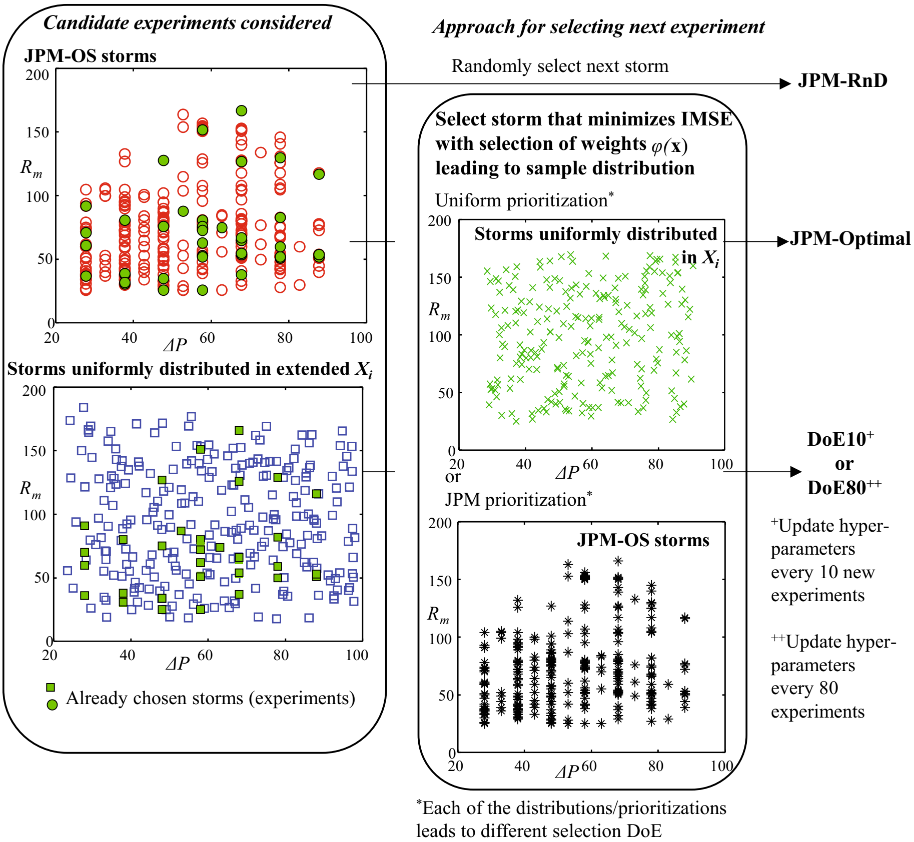

Fig. 4 Demonstration of different DoE approaches considered, along with the two different approaches for *IMSE* definition. Projection to ΔP and R_m inputs is utilized for demonstration

of the box-bounded domain for each input dimension. Since the surrogate model will be ultimately used to provide predictions within X, this extension was deemed necessary to mitigate the boundary effect, since the surrogate model accuracy is expected to decrease drastically at the boundary of the domain used in its development. The third approach, termed DoE10, updates the surrogate model hyper-parameters after addition of 10 new storms, and the fourth approach, termed DoE80, updates the surrogate model hyper-parameters after adding 80 new storms. This fourth approach is motivated by the fact that numerical simulation of storms is frequently performed in larger batches to exploit parallelization efficiency. Update of the surrogate model also involves update of the PCA, retaining always sufficient number of latent outputs to account for 99.9% of the variability of the actual output $\mathbf{y}$. This choice (i.e., the 99.9% selection) guarantees that any errors stemming from the PCA approximation are negligible; therefore, the accuracy statistics only represent errors stemming from the surrogate model approximation.

For the *IMSE* calculation, the domain X^I corresponds to X and two cases are examined for $\varphi(\mathbf{x})$ using either uniform distribution with $N_q = 1000$ samples, or the JPM-OS selection of the 595 storms ($\mathbf{x}^q = \mathbf{x}^l$). The weight w_q (see Appendix 3 for definition) for the latter case corresponds to the sum of the frequency weights of storms

$$w_q(\mathbf{x}^l) = \sum_{j=1}^{n_y} \lambda_j^l \tag{16}$$

where as defined earlier, λ_j^l denotes the frequency weight (mean annual rate) for landfalling storm l for save location j. These cases will be referred to, respectively, as "uniform DoE prioritization" and "JPM DoE prioritization," and are also summarized in Table 1 and illustrated in Fig. 4. The test-sample used in validation for each case is taken to have the same characteristics as $\varphi(\mathbf{x})$ for consistency in comparisons: This means 1000 uniformly distributed samples in X for "uniform DoE prioritization" and the 595 JPM-OS storms for the "JPM DoE prioritization." For the latter case (JPM DoE prioritization), if any test-sample storm happens to belong to the database used to develop the Kriging model, its prediction errors are evaluated using the LOOCV approach.

For the optimization of Eq. (14) n_c is chosen as 10,000 candidate experiments, whereas the reduction factors a_r and a_c (discussed in Appendix 3) are both set, respectively, equal to 0.5 to reduce computational effort. Sensitivity analysis performed showed that this value did not impact the quality of the identified solutions. All approaches initiate from a common (to facilitate consistent comparisons) set of 40 storms, randomly chosen from the initial 595 storms. For the *IMSE*-based optimization approaches this initialization is necessary for building the surrogate model used to identify the next best storm to add in the database. Figures 3 and 4 show the improvement of Kriging surrogate model accuracy as the number of experiments increases. As error statistics the average correlation coefficient $\overline{cc}$, coefficient of determination $\bar{R}^2$ and root-mean-square error $\overline{rmse}$ are utilized. Figure 5 shows results using test-sample, and Fig. 6 using LOOCV validation. In each case two different statistics are shown (compromise due to space limitations) $\overline{cc}$ and $\overline{rmse}$ in Fig. 6 and $\overline{cc}$ and $\bar{R}^2$ in Fig. 6. This allows to show the similar behavior with respect to the different statistics and at the same time do a direct comparison between the two cases (since both use $\overline{cc}$). For the DoE80 points, rather than continuous curves, single points are reported since validation of accuracy is performed only after 80 new experiments. Figure 7 then shows the risk predictions for landfalling storms DoE10 and RnD10 for the two different approaches for definition of $\varphi(\mathbf{x})$, for values of n = 50, 90, and 190. Risk estimates are obtained by using the surrogate model to provide approximation to the 595 JPM-OS storms and then calculating risk based on the JPM-OS weights as shown in Eq. (15).

Comparing first the trends for the different error statistics, it is evident that they are very similar. Hence, monitoring any single one would be sufficient, as its improvement would indicate improvement for the others. For this reason, the discussion focuses herein on the correlation coefficient $\overline{cc}$. Comparison between the LOOCV (Fig. 6) and test-sample validation (Fig. 5) shows that the former significantly overestimates the accuracy for small n values. This is a well-known limitation of cross-validation approaches for small data sets (Rao et al. 2008). Ultimately for small values of n (smaller than 10–20 times the dimension of input) the available hurricane scenarios to perform the LOOCV (recall cross-validation only uses the n experiments within the database) are not sufficient for reliably estimating accuracy statistics. This is also the reason why for some cases in Fig. 6 the initial increase in n decreases the LOOCV accuracy statistics; this does not mean that the addition of experiments has reduced the quality of Kriging, simply that the accuracy for LOOCV statistics has improved, leading to smaller overestimation and, ultimately, to decrease in

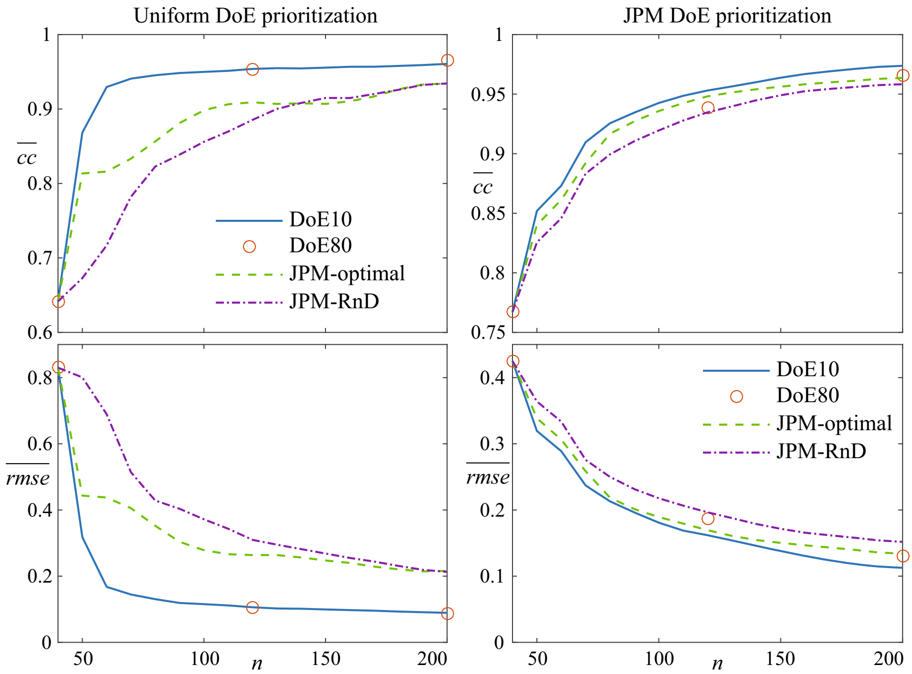

Fig. 5 Improvement of surrogate model accuracy with n for average correlation coefficient (top row) and root-mean-square error (bottom row), for the four different storm selection approaches for validation using test-sample approach. Results for both uniform DoE prioritization (left column) and JPM DoE prioritization (right column) are shown

these statistics in later iterations. Test-sample approach, using a large number for hurricane scenarios for validating accuracy (independent of n value), does not suffer from this critical shortcoming of potential overestimation of prediction accuracy and is therefore chosen as the validation approach for evaluating the DoE efficiency. It should be noted, though, that for larger n values the LOOCV and test-sample statistics converge. In other words, there should be no reservation in relying on LOOCV statistics for measuring surrogate model accuracy when the database is large enough. Nevertheless, this discussion shows that for evaluating DoE efficiency one should have in mind that the LOOCV statistics may offer an overestimation when the database size n is still small. The degree of this overestimation (how large it is in other words) is related to the specifics of the database and the application at hand (Rao et al. 2008). A possible remedy for this issue could be to increase the number of storms left out for validation purpose (for example, the k-fold CV described in Appendix 1). A larger number of left-out storms can potentially lower the CV error variance and mitigate the overestimation, because it reduces the overlap between Kriging models fitted on retained storms. It might, though, have the opposite effect and increase the CV error bias because of the decreased size of retained storms. Hence, it has been acknowledged that the effect of the left-out storm number on the CV performance is application and data dependent (Bengio and Grandvalet 2004), and a dedicated selection of this number to the problem at hand is beyond current paper's scope.

Moving now to the comparison across the DoE approaches, the explicit optimization for the storm to add can yield considerable improvement in Kriging surrogate model accuracy

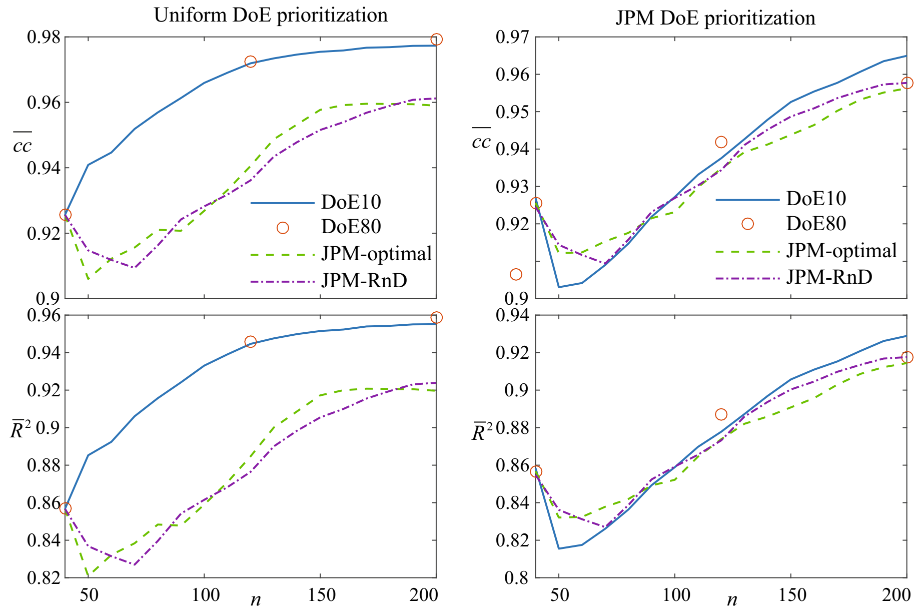

Fig. 6 Improvement of surrogate model accuracy with n for average correlation coefficient (top row) and coefficient of determination (bottom row), for the four different storm selection approaches for LOOCV. Results for both uniform DoE prioritization (left column) and JPM DoE prioritization (right column) are shown

(compare JPM-RnD to other three approaches), whereas there can be distinct benefits in searching for such a storm in a continuous domain D rather than choosing among synthetic storms with predetermined discretized characteristics (compare DoE and JPM-Optimal cases). As n increases there is an initial steep performance improvement, which gradually reaches a plateau, as sufficient experiments are already added to establish high-accuracy predictions. This indicates that for gradual enrichment of the observation database, a reasonable termination criterion is to monitor the Kriging accuracy so that the process can stop when either the desired accuracy threshold has been reached (as discussed in Sect. 2.4) or the improvement trend enters a plateau. Beyond that point, the addition of observations offers a smaller marginal improvement. Based on the discussion in the previous paragraph, care should be taken when using LOOCV to evaluate this accuracy improvement if the number of storms in the database n is still moderately small, in order to avoid premature termination relying on overestimated accuracy characteristics. This can be facilitated by adopting such criteria only if n exceeds a minimum threshold (20 times the dimension of the input should be in general sufficient). Comparing, now, RnD and JPM-Optimal cases it is evident that they quickly converge even for the uniform DoE prioritization where big difference in initial iterations exists. This is anticipated; since RnD and JPM-Optimal select storms from the same set, as n increases the databases utilized to form the surrogate models gradually share greater similarities (these databases will end up being completely identical when $n = 595$), which then leads to similarities in the predictive abilities of these surrogate models and ultimately to a smaller marginal improvement by the selection of the best new storm.

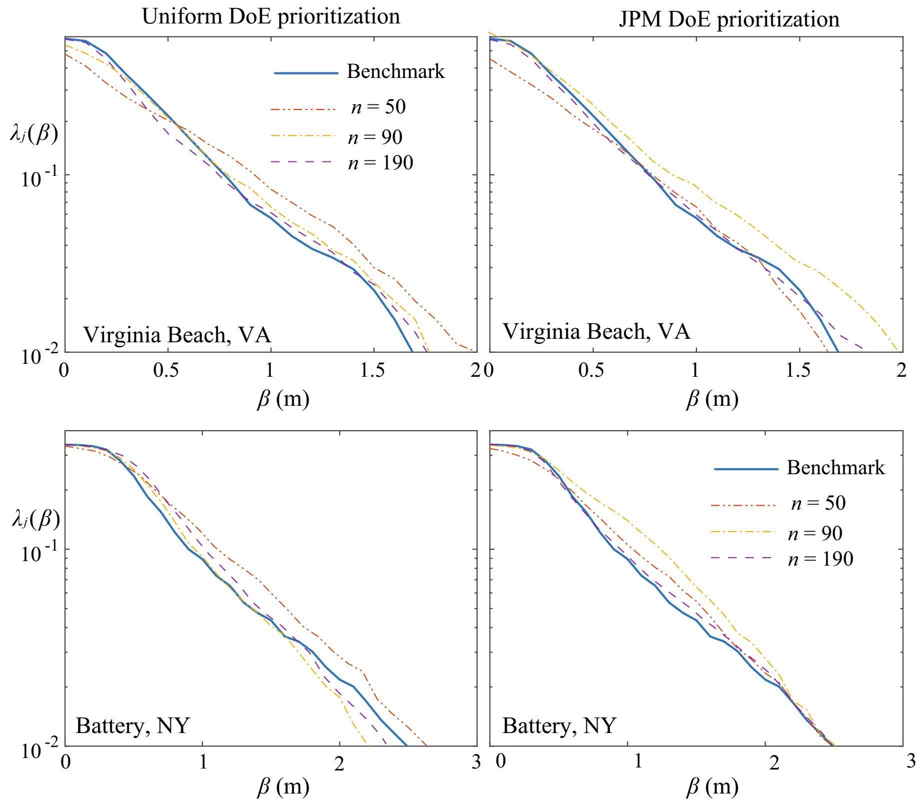

Fig. 7 Risk estimates [annual exceedance rate $\lambda_j(\beta)$ for threshold β] for landfalling storms for Virginia Beach, VA (Virginia), and The Battery, NY (New York), locations using DoE10 for (n = 50, 90 or n = 180) and comparison to benchmark risk estimates. Results for both uniform DoE prioritization (left column) and JPM DoE prioritization (right column) are shown

The selection of $\varphi(\mathbf{x})$, in other words of the subdomains in the input space to provide higher priority to, has an important effect on the performance of different DoE approaches (compare the two columns in Fig. 5 or even Fig. 6). If all storms are prioritized equally (uniform DoE prioritization case), the benefits from the explicit DoE optimization are more substantial (compare DoE and JPM-Optimal cases), whereas how many storms to add in a single batch (compare DoE10 to DoE80) does not have a strong influence. On the other hand, prioritizing storms according to JPM-OS (JPM DoE prioritization case) weakens benefits from the adaptive DoE scheme, whereas in this case adding larger number of storms in a single batch results in significantly worse performance. This should be attributed to the characteristics of the JPM database, distributed in specific subdomains within the greater domain X (the histogram of input values in Fig. 4 clearly demonstrates this). Due to this characteristic, even choosing new experiments randomly in these subdomains (JPM-RnD case) provides sufficient benefits that the explicit optimization for the best storm cannot outperform by a large margin. This discussion demonstrates the importance for carefully formulating the database enrichment approach by selecting an *IMSE* weight $\varphi(\mathbf{x})$ that is appropriate for the intended use of the surrogate model, either that is facilitating accurate predictions over the entire domain of interest or that is focusing on specific subdomains. It also shows that the benefits for the adaptive database enrichment will not always be as substantial, with its best performance occurring for the uniform prioritization case. In that instance, the addition of a larger number of storms has only a small influence on the potential surrogate model improvement. This is not, though, necessarily true for all potential $\varphi(\mathbf{x})$ selections.

The comparisons with respect to risk in Fig. 7 validate the aforementioned trends. As the number of storms in the database n increases, the accuracy of the risk predictions supported through Kriging improves and converges to the benchmark risk. The improvement is greater when the JPM DoE prioritization is utilized. This is anticipated: when the surrogate model is used for a specific set of storm (the ones in the risk assessment here), then selection of the experiments with an explicit goal to improve accuracy over this set will help in providing higher-quality risk estimates. This again stresses the importance of the proper selection of $\varphi(\mathbf{x})$. Note that for the estimation of risk integral of Eq. (15) using Kriging, the surrogate model prediction error can be explicitly incorporated in the analysis (Jia and Taflanidis 2013). This was not done here so that the analysis depicts directly the differences that stem from the Kriging predictions, given by Eq. (2), at every examined n value, that is, in order to avoid integrating in the risk estimates additional differences that may stem from the prediction error. To further investigate the trends identified in Fig. 7, Fig. 8 evaluates the surrogate model accuracy for different sizes n of the database using a test-sample with different characteristics than the prioritization adopted for the DoE. As such for uniform DoE prioritization the set of 595 JPM storms is used as test-sample set, whereas for the JPM DoE prioritization the set of 1000 storm uniform test-sample is used. Results show that improvement of global surrogate model accuracy might come at the expense of reduced improvement for a specific storm set (check the reduction in prediction quality for the DoE cases adopting a uniform DoE prioritization when evaluated strictly over the 595 JPM storms). This trend is aligned with the better risk predictions offered by the JPM DoE prioritization previously shown in Fig. 7.

Overall, the illustrative application in this section demonstrates that substantial computational benefits can be accomplished by adaptively selecting the database of synthetic storms informing the Kriging metamodel: The same level of accuracy can be established

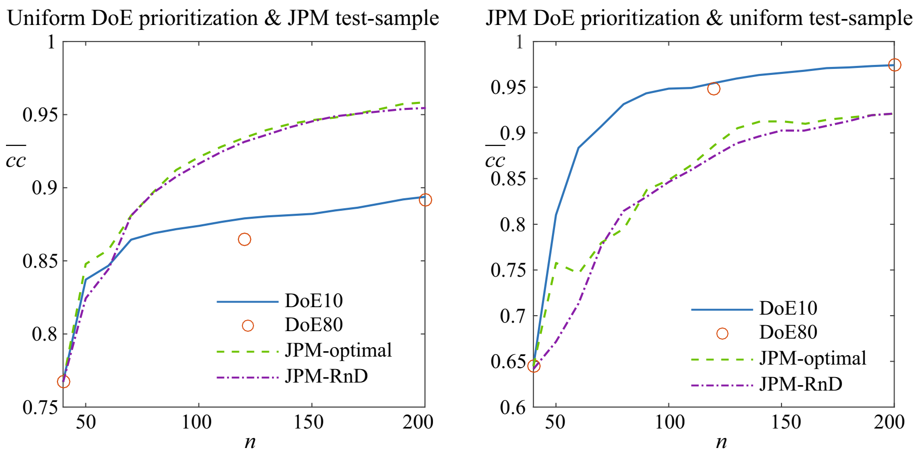

Fig. 8 Evaluation of surrogate model accuracy for different sizes of the observation database n. The test-sample used for the validation has different characteristics with the prioritization adopted in the DoE

using significantly smaller number of storms when an adaptive selection is applied, using the weighted metamodel predicted error to guide the selection. The magnitude of the computational benefit depends on the intended use of the metamodel, whether accuracy is desired over a broader range of future storms characteristics or simply for a specific set of storms, a priori known. Benefits are substantially larger in the former case, which is especially important when the surrogate model is intended to be used for emergency response management applications. The formulation of the adaptive DoE, specifically the appropriate prioritization of storm characteristics when calculating the weighted error, needs to appropriately reflect this intended use.

5 Addressing characteristics related to climate change: storm intensification and sea level rise

5.1 Addressing storm intensification

The next topic investigated is the impact of storm intensification. The initial database is split into two groups: 405 lower-intensity storms with $\Delta P < 65$ hPa and 190 higher-intensity storms with $\Delta P > 65$ hPA. The corresponding subdomains in X are denoted as X^l ($\Delta P \in$[28 65] hPa) and X^h ($\Delta P \in$[66 98] hPa), respectively. For the purposes of this investigation the latter are assumed to correspond to scenarios representing future intensification of storms. The surrogate model is developed using only the storms in X^l (i.e., $X = X^l$), and its accuracy is evaluated separately for the storms in X^l (using LOOCV) and the storms in X^h (using test-sample validation). The latter represents the extrapolation prediction accuracy for storms having intensified characteristics. The former represents the interpolation prediction accuracy within the domain encompassed by the observation database. Both MLE and CV are considered for the hyper-parameter tuning, whereas three cases are examined: no weights $w_h = 1$ and the introduction of higher weights for the storms (within X^l) with higher intensity, choosing $w_h = \Delta P^2$ or ΔP^4. For the two weighted cases the effect of adding 10 storms is also examined, selected through the approach outlined in Sect. 2.4, and targeting only the domain representing intensified storms, with D constrained to be X^h. Accuracy results for all cases are presented in Table 2 with respect to the average coefficient of determination, correlation coefficient and root-mean-square error. Figure 9 shows for some of the cases the correlation coefficient per storm as a function of ΔP. The statistics for each storm are calculated by modifying Eq. (19) [discussed in Appendix 1] to estimate statistics per storm (instead of statistics per save point):

$$cc_h = \frac{\sum_{j=1}^{n_y}\left(y_j(\mathbf{x}^h) - \bar{y}_h\right)\left(\hat{y}_j(\mathbf{x}^h|\mathbf{X}_{-h}) - \bar{\hat{y}}_h\right)}{\sqrt{\sum_{j=1}^{n_y}\left(y_j(\mathbf{x}^h) - \bar{y}_h\right)^2 \sum_{j=1}^{n_y}\left(\hat{y}_j(\mathbf{x}^h|\mathbf{X}_{-h}) - \bar{\hat{y}}_h(\mathbf{X}_{-h})\right)^2}}$$
$$\text{where } \bar{\hat{y}}_h(\mathbf{X}_{-h}) = \frac{1}{n_y}\sum_{j=1}^{n_y}\hat{y}_j(\mathbf{x}^h|\mathbf{X}_{-h}), \bar{y}_h = \frac{1}{n_y}\sum_{j=1}^{n_y}y_j(\mathbf{x}^h) \tag{17}$$

With respect to the tuning of hyper-parameters, first, it is clear from the results in Table 2 that for extrapolation accuracy CV outperforms MLE by a considerable margin across all cases. This trend agrees with observation from similar studies (Ginsbourger et al. 2009; Bachoc 2013) and may be attributed to overfitting enforced by MLE. CV offers enhanced accuracy both for extrapolation and for interpolation, though for the latter with only a small margin. This margin decreases further as the surrogate model accuracy

Table 2 Accuracy (average coefficient of determination, correlation coefficient, and root-mean-square error) for different scenarios related to storm intensification study

Optimization approach	No weights		With weights ΔP^2		With weights ΔP^2 and 10 added storms		With weights ΔP^4		With weights ΔP^4 and added storms	
	MLE	CV	MLE	CV	MLE	CV	MLE	CV	MLE	CV
X^l (interpolation)										
$\bar{R}^2$	0.931	0.945	0.928	0.944	0.932	0.944	0.934	0.944	0.923	0.939
$\overline{cc}$	0.961	0.972	0.958	0.972	0.967	0.971	0.966	0.971	0.961	0.969
$\overline{rmse}$ (m)	0.100	0.081	0.101	0.085	0.093	0.083	0.095	0.085	0.103	0.090
X^h (extrapolation)										
$\bar{R}^2$	0.743	0.890	0.765	0.912	0.854	0.945	0.842	0.919	0.869	0.946
$\overline{cc}$	0.892	0.958	0.899	0.966	0.937	0.975	0.939	0.968	0.945	0.975
$\overline{rmse}$ (m)	0.360	0.232	0.344	0.205	0.267	0.158	0.277	0.195	0.254	0.157

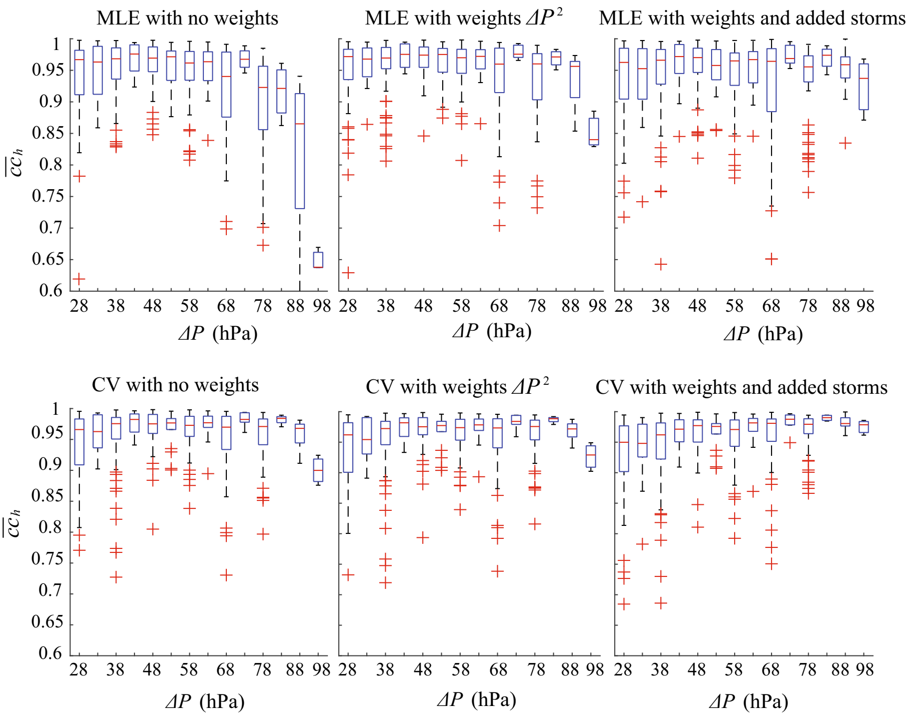

Fig. 9 Accuracy (average correlation coefficient) per storm as a function of the intensity ΔP for some of the surrogate model tuning approaches examined for the scenarios evaluating the impact of storm intensification. Hyper-parameter optimization using both MLE (top row) and CV (bottom row) is shown. Results are presented as box plot, with boxes indicating 25% and 75% sample quantile, red line the median, crosses outliers, and dashed lines the last non-outlier sample

improves (compare the results reported here to the results in Sect. 3 when using the entire storm database). These trends indicate a clear preference for hyper-parameter optimization using CV approach.

Moving now to the introduction of the intensity-related weights, the results in Table 2 and Fig. 9 show that this introduction substantially enhances the extrapolation prediction accuracy with only a small sacrifice of the interpolation accuracy. Using higher order of weights (ΔP^4 instead of ΔP^2) provides greater extrapolation benefits, though, at the cost of a bigger reduction in interpolation quality. This indicates that the impact of weight selection on the interpolation accuracy needs to be carefully examined through a parametric sensitivity study. Note that in practical implementation this (i.e., the interpolation accuracy evaluation) will be the only available information since a test-sample to validate extrapolation accuracy is not readily available. The addition of (10 in this case) new, high-intensity storms further improves the extrapolation accuracy. This is especially evident in Fig. 9 (compare the third column to the first or second one).

Overall, the illustrative application in this section demonstrates the potential of improving accuracy of the Kriging metamodel when used for predictions of storms outside the initial storm parameter space, more specifically with intensified characteristics compared to the database of synthetic storms used for its development through: (1) the proper tuning of the metamodel, introducing appropriate weights in the hyper-parameter optimization and selecting CV as optimization objective, and, if possible, (2) enrichment of database with new storms. The aforementioned (extrapolation) accuracy improvement can be accomplished with only small impact to the (interpolation) predictive capabilities when the metamodel is used to provide predictions for storms with characteristics that fall within the range of the initial database.

5.2 Addressing sea level rise

The final topic investigated is the ability to evaluate SLR impact using Kriging. For this investigation a separate Kriging model is trained to evaluate the difference between Base + Tides + SLR and Base + Tide scenarios in the NACCS database. This difference represents the impact of the sea level rise and is denoted as SLR herein. Since in the NACCS only one SLR scenario was explicitly examined through hydrodynamic simulation, the analysis is restricted to treating SLR as a different output, rather than incorporating it as characteristic of the input vector **x**. Tuning of Kriging to directly predict the output for Base + Tides + SLR was found to have low accuracy, presumably due to the effect of the interaction between storm surge and random phase tides. Since, for the purpose of this study, no information on specific tidal phase is used, it cannot be directly used as surrogate model input. By subtracting the results for the Base + Tides + SLR and Base + Tide scenarios the impact of the random tides is minimized, allowing for the development of a Kriging model to directly predict the impact of SLR on different hurricane scenarios.

The Kriging surrogate model developed for the SLR output has, again, good accuracy with average correlation coefficient 96.8%, coefficient of determination 90.5% and a root-mean-square error 0.0343 m. The surrogate model is then utilized to provide risk estimate predictions for the Base + SLR scenario as shown in Fig. 10. Predictions for both Base and Base + SLR scenarios are shown with their benchmark (using the actual database) and

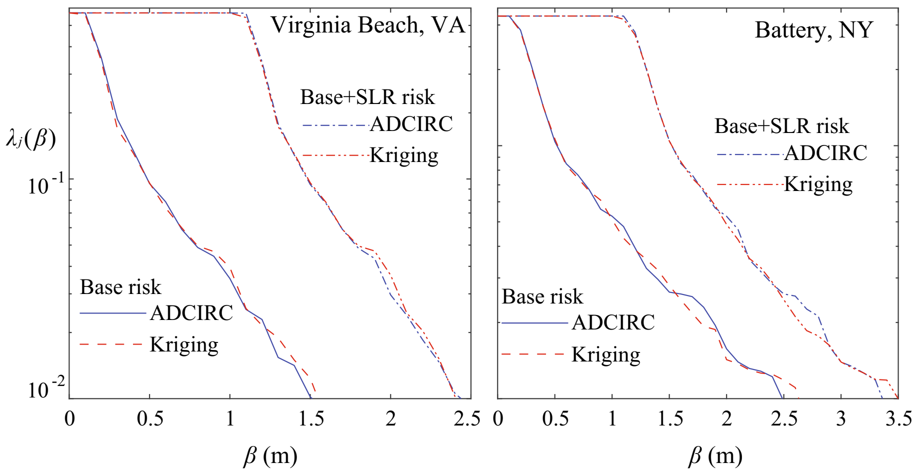

Fig. 10 Risk estimates [annual exceedance rate $\lambda_j(\beta)$ for threshold β] for Virginia Beach, VA (Virginia), and The Battery, NY (New York), locations with and without SLR consideration. Kriging-based estimates and estimates using initial simulations (ADCIRC) are shown

Kriging-based estimates. The Kriging estimates are established using LOO predictions since the scenarios examined belong to the database. The high accuracy of the surrogate models for both the Base and SLR scenarios is validated from the results in the figure, as Kriging and benchmark (ADCIRC) risk curves have very small differences. Overall, the illustrative application in this section demonstrates the potential to accurately predict SLR impact using Kriging metamodeling.

6 Conclusions

Advancements were developed for the application of Kriging surrogate modeling for storm surge predictions with respect to: (a) the exact selection of the synthetic storms for creating the database that will inform the surrogate model development and (b) adjustments to support two implementation issues that might become more relevant due to climate change considerations, prediction for intensified storms compared to the ones available in initial synthetic-storm database, and surge estimation that considers sea level rise (SLR) impact. All advances were demonstrated using the database for the North Atlantic Comprehensive Coastal Study (NACCS) recently developed by the US Army Corps of Engineers.

For storm selection an adaptive, sequential DoE was examined that iteratively identifies the storm (or multiple storms) whose addition to the database is expected to provide the greatest benefit. The weighted integrated mean squared error (*IMSE*) was used as the objective function for selecting the next best storm, and a computationally efficient implementation was discussed for the associated optimization. Results demonstrated the potential benefits of the adaptive DoE for providing improved surrogate model accuracy and higher-quality predictions for surge risk. They also stressed the importance for properly formulating the DoE problem, selecting weights for the *IMSE* function that appropriately prioritize storm characteristics (i.e., subdomains in the input domain) that coincide with the intended use of the surrogate model, that corresponds to either a global accuracy over a range of candidate storms (this might be important for use as a tool for managing emergency responses during landfalling events) or an improved accuracy for a specific set of storms (this is important for risk estimation using JPM principles). It was also shown that using leave-one-out cross-validation (LOOCV) for evaluating surrogate model quality in this context should be carefully used, as it might overestimate prediction accuracy when the size of database is still relatively small. Since this accuracy might be used as a measure for evaluating if new storms are needed (to facilitate further improvement), care should be taken to avoid premature termination of the adaptive DoE due to overestimated accuracy characteristics.

Storm intensification can be addressed by weighing the different storms in the existing database, giving larger weights to storms of higher intensity. This facilitates greater extrapolation accuracy for future, even stronger storms. In this context, the Kriging model hyper-parameter optimization based on predictive capabilities (cross-validation) was shown to offer significant advantages over the common approach using maximum likelihood estimation. Further improvement of extrapolation capabilities can be facilitated by a database enrichment strategy utilizing adaptive DoE. Finally, it was shown that Kriging surrogate modeling can be used to accurately support surge predictions considering SLR.

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Appendix 1: Surrogate model validation

Validation of the surrogate model is established by comparing its predictions to the actual response (high-fidelity simulations of storm surge) for storm scenarios that do not belong to database $\mathbf{X}$. These scenarios represent the validation set. Two common approaches are used for forming this set (Kohavi 1995): test-sample or cross-validation. The first one requires new observations, or, equivalently, splitting of the initial database to a single training set and a single validation set. The second approach repeats comparisons of the accuracy over different divisions of the entire database to training set and validation set and has the benefit that it requires no new observations. Perhaps the most common implementation of the latter is the leave-one-out cross-validation (LOOCV): Each of the observations from the database is sequentially removed, then the remaining training points are used to predict the output for it, and the error between the predicted and real responses is evaluated. A more general CV approach is k-fold CV, established by partitioning the entire observation set into k equal-size subsets and calculating errors similarly as in the LOOCV, but removing each time the entire subset, rather than a single observation.

As error statistics popular choices are the coefficient of determination, the correlation coefficient and the square root of the mean squared error. For output y_j and the LOOCV approach these are given by

$$R_j^2 = 1 - \frac{\sum_{h=1}^{n}\left(y_j(\mathbf{x}^h) - \hat{y}_j(\mathbf{x}^h|\mathbf{X}_{-h})\right)^2}{\sum_{h=1}^{n}\left(y_j(\mathbf{x}^h) - \bar{y}_j\right)^2};\ \bar{y}_j = \frac{1}{n}\sum_{h=1}^{n} y_j(\mathbf{x}^h) \tag{18}$$

$$cc_j = \frac{\sum_{h=1}^{n}\left(y_j(\mathbf{x}^h) - \bar{y}_j\right)\left(\hat{y}_j(\mathbf{x}^h|\mathbf{X}_{-h}) - \bar{\hat{y}}_i\right)}{\sqrt{\sum_{h=1}^{n}\left(y_j(\mathbf{x}^h) - \bar{y}_j\right)^2 \sum_{h=1}^{n}\left(\hat{y}_j(\mathbf{x}^h|\mathbf{X}_{-h}) - \bar{\hat{y}}_j(\mathbf{X}_{-h})\right)^2}};\ \bar{\hat{y}}_j(\mathbf{X}_{-h}) = \frac{1}{n}\sum_{h=1}^{n} \hat{y}_j\left(\mathbf{x}^h|\mathbf{X}_{-h}\right) \tag{19}$$

$$rmse_j = \sqrt{\frac{1}{n}\sum_{h=1}^{n}\left(y_j(\mathbf{x}^h) - \hat{y}_j(\mathbf{x}^h|\mathbf{X}_{-h})\right)^2}, \tag{20}$$

where $\hat{y}_j(\mathbf{x}^h|\mathbf{X}_{-h})$ represents the surrogate model predictions for storm $\mathbf{x}^h$ established by using the entire database excluding that storm. Using the closed form of Eq. (7) the statistics in Eqs. (18)–(20) can be calculated without explicitly evaluating $\hat{y}_j(\mathbf{x}^h|\mathbf{X}_{-h})$, rather simply substituting LOO predictions for entire output vector by

$$\hat{\mathbf{y}}\left(\mathbf{x}^h|\mathbf{X}_{-h}\right) = \mathbf{P}\left(\mathbf{z}(\mathbf{x}^h) + [\mathbf{e}]_h^{\mathrm{T}}\right), \tag{21}$$

where $[\mathbf{e}]_h$ is the vector corresponding to the hth row of matrix with elements e_{hi}, i.e., to the LOO errors associated with the latent outputs for storm $\mathbf{x}^h$. For the test-sample approach the set $\{\mathbf{x}^h; h = 1, \ldots, n\}$ needs to be replaced with the training set of new observations $\{\mathbf{x}_n^h; h = 1, \ldots, n_n\}$ in these equations, whereas the entire database $\mathbf{X}$ (rather than $\mathbf{X}_{-h}$) is used for the surrogate model predictions. It should be pointed out that the aforementioned statistics evaluate the average accuracy over the examined domain $\mathbf{X}$. An alternative approach would have been to look at the maximum error over $\mathbf{X}$ which examines the worst-case prediction performance.

The accuracy across all outputs is evaluated by calculating the average error statistics over all outputs.

$$\bar{R}^2 = \frac{1}{n_y}\sum_{j=1}^{n_y} R_j^2; \quad \overline{cc} = \frac{1}{n_y}\sum_{j=1}^{n_y} cc_j; \quad \overline{rmse} = \frac{1}{n_y}\sum_{j=1}^{n_y} rmse_j. \tag{22}$$

Values of $\bar{R}^2$ and $\overline{cc}$ close to 1 and values of $\overline{rmse}$ close to zero indicate better accuracy. It should be also pointed out that $\overline{rmse}$ does not correspond to normalized statistics; in other words, its value depends on the magnitude of the response.

Appendix 2: Calculation of gradients for LOOCV objective function

This appendix discusses the estimation of the gradient of the objective function of Eq. (6) utilized in the LOOCV tuning of the hyper-parameters. The kth component of the gradient vector, the partial derivative with respect to hyper-parameter s_k, is

$$\frac{\partial H_m(\mathbf{s})}{\partial s_k} = \frac{2}{m_c}\sum_{i=1}^{m_c}\gamma_i\left(\frac{1}{n}\sum_{h=1}^{n} e_{hi}\frac{\partial e_{hi}}{\partial s_k}\right) \tag{23}$$

with partial derivative for LOO error given by

$$\frac{\partial e_{hi}}{\partial s_k} = \frac{\partial (g_{hi}/c_{hh})}{\partial s_k} = \left(c_{hh}\frac{\partial g_{hi}}{\partial s_k} - g_{hi}\frac{\partial c_{hh}}{\partial s_k}\right)/(c_{hh})^2, \tag{24}$$

where

$$\begin{aligned}\frac{\partial g_{hi}}{\partial s_k} &= -\left[\mathbf{R}(\mathbf{X})^{-1}\cdot\frac{\partial \mathbf{R}(\mathbf{X})}{\partial s_k}\cdot\mathbf{R}(\mathbf{X})^{-1}(\mathbf{Z}-\mathbf{F}(\mathbf{X})\boldsymbol{\beta}^*) + \mathbf{R}(\mathbf{X})^{-1}\mathbf{F}(\mathbf{X})\frac{\partial \boldsymbol{\beta}^*}{\partial s_k}\right]_{hi} \\ \frac{\partial c_{hh}}{\partial s_k} &= -\mathbf{R}(\mathbf{X})^{-1}\left[\frac{\partial \mathbf{R}(\mathbf{X})}{\partial s_k}\right]_{hh}\mathbf{R}(\mathbf{X})^{-1},\end{aligned} \tag{25}$$

in which $\frac{\partial \mathbf{R}(\mathbf{X})}{\partial s_k}$ and $\frac{\partial \boldsymbol{\beta}^*}{\partial s_k}$ are matrices of element-wise derivatives with the latter given by

$$\frac{\partial \boldsymbol{\beta}^*}{\partial s_k} = \left(\mathbf{Q}(\mathbf{X})^{\mathrm{T}}\mathbf{F}(\mathbf{X})\right)^{-1}\mathbf{Q}(\mathbf{X})^{\mathrm{T}}\cdot\frac{\partial \mathbf{R}(\mathbf{X})}{\partial s_k}\cdot\left(\mathbf{Q}(\mathbf{X})\left(\mathbf{Q}(\mathbf{X})^{\mathrm{T}}\mathbf{F}(\mathbf{X})\right)^{-1}\mathbf{F}(\mathbf{X})^{\mathrm{T}} - \mathbf{I}\right)\mathbf{R}(\mathbf{X})^{-1}\mathbf{Z} \tag{26}$$

with $\mathbf{Q}(\mathbf{X})^{\mathrm{T}} = \mathbf{F}(\mathbf{X})^{\mathrm{T}}\mathbf{R}(\mathbf{X})^{-1}$.

Appendix 3: Optimization for storm selection

This appendix discusses a numerical optimization scheme for the storm selection. First, select the number of candidate samples n_c for the stochastic search, the percentage reductions a_r, a_c for determining the retained experiments, and the number of samples for the Monte Carlo integration (MCI) N_s. Perform then the following steps, also shown in Fig. 11 for the same example considered in Fig. 1.

(1) [MCI samples]. Generate N_s $\{\mathbf{x}^q;\ q = 1,\ldots, N_s\}$ samples representing distribution $\varphi(\mathbf{x})$, and determine the corresponding weights $w_q(\mathbf{x}^q)$. For the common implementation that samples are directly from distribution $\varphi(\mathbf{x})$ the weights are $w_q(\mathbf{x}^q) = 1$.

(2) [Candidate experiment generation]. Generate n_c samples for $\mathbf{x}$ with uniform distribution in domain D, forming set $\{\mathbf{x}^c_{new}; c = 1,\ldots,n_c\}$. Skip this step if candidate experiments are already provided (Part a of Fig. 11).

(3) [Ranking of experiments]. Evaluate $\{\sigma_n^2(\mathbf{x}^c_{new}|\mathbf{X}); c = 1,\ldots,n_c\}$ and order candidate samples using a descending order for $\sigma_n^2(\mathbf{x}^c_{new}|\mathbf{X})$. Retain only the $a_r n_c$ candidate experiments that correspond to the highest values of $\{\sigma_n^2(\mathbf{x}^c_{new}|\mathbf{X}); c = 1,\ldots,n_c\}$ (Part b of Fig. 11). Skip this step if $a_r = 1$.

(4) [Clustering of solutions]. Cluster retained experiments from Step 3 into $a_c a_r n_c$ clusters, using for example K-means cluster (Hartigan and Wong 1979), and keep only one experiment per cluster, the one corresponding to the largest value of $\sigma_n^2(\mathbf{x}^c_{new}|\mathbf{X})$ (Part c of Fig. 11). Skip this step if $a_c = 1$.

(5) [Calculation of *IMSE*]. For each retained candidate experiment after Step 4, $\{\mathbf{x}^c_{new}; c = 1,\ldots,a_r a_c n_c\}$, calculate $\{\sigma_n^2(\mathbf{x}^q|\mathbf{X},\mathbf{x}^c_{new}); q = 1,\ldots,N_q\}$ using Eqs. (12) and (13). Approximate *IMSE* through MCI as

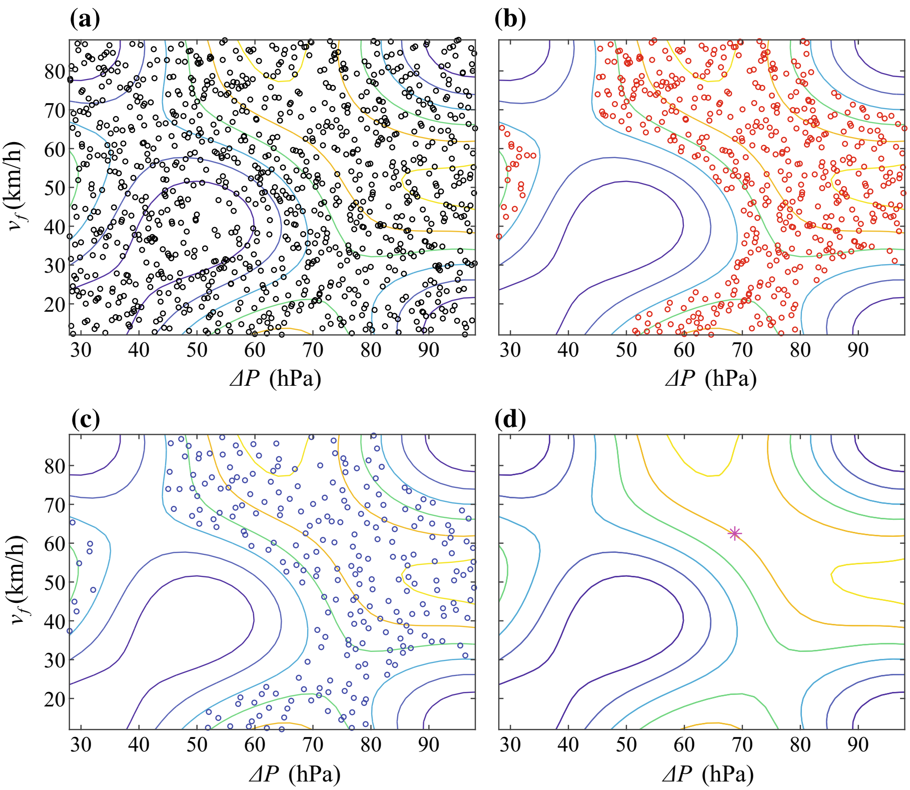

Fig. 11 Demonstration of the different steps for the numerical implementation of the adaptive DoE for the same two-dimensional example shown in Fig. 1. In all the curves the contours correspond to the normalized variance. **a** Initial candidate experiments; **b** experiments retained after a 50% ranking based on [$a_r = 0.5$]; **c** experiments retained after 50% clustering [$a_c = 0.5$]; **d** final experiment obtained after minimization of *IMSE*

$$IMSE(\mathbf{x}^c_{new}) = \frac{1}{N_s}\sum_{q=1}^{N_s} w_q(\mathbf{x}^q)\sigma_n^2(\mathbf{x}^q|\mathbf{X}, \mathbf{x}^c_{new}). \tag{27}$$

(6) [Final selection]. Select as new experiment the one that provides the minimum value for $\{IMSE(\mathbf{x}^c_{new}); c = 1, \ldots, a_r a_c n_c\}$ (Part d of Fig. 11).

The most computationally demanding step of this process is Step 5, which requires MCI for each of the candidate experiments examined. Steps 3 and 4 have been introduced to reduce this burden. Step 3 removes candidate experiments belonging to domains where the surrogate model accuracy is already high. For such experiments, it is anticipated that their *IMSE* will not correspond to a minimum over *D*, as addition of experiments in domains of already adequate accuracy [corresponding to lower values of $\sigma_n^2(\mathbf{x}|\mathbf{X})$] is anticipated to provide small benefits. Step 4 keeps only one candidate experiment among experiments that are close to one another. All such experiments are expected to provide similar improvement of *IMSE*, and therefore, examining all of them as candidate solutions is redundant. Only the experiment with lowest accuracy is retained in each cluster. Removal of experiments in Steps 3 and 4 provides an $a_c a_r$-fold reduction in computational burden without compromising the quality of the identified solution of the optimization, provided that values of a_r and a_c are not too small. These values should be chosen in range [0.7–0.3].

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